Global Knowledge®

# Angular
## Services

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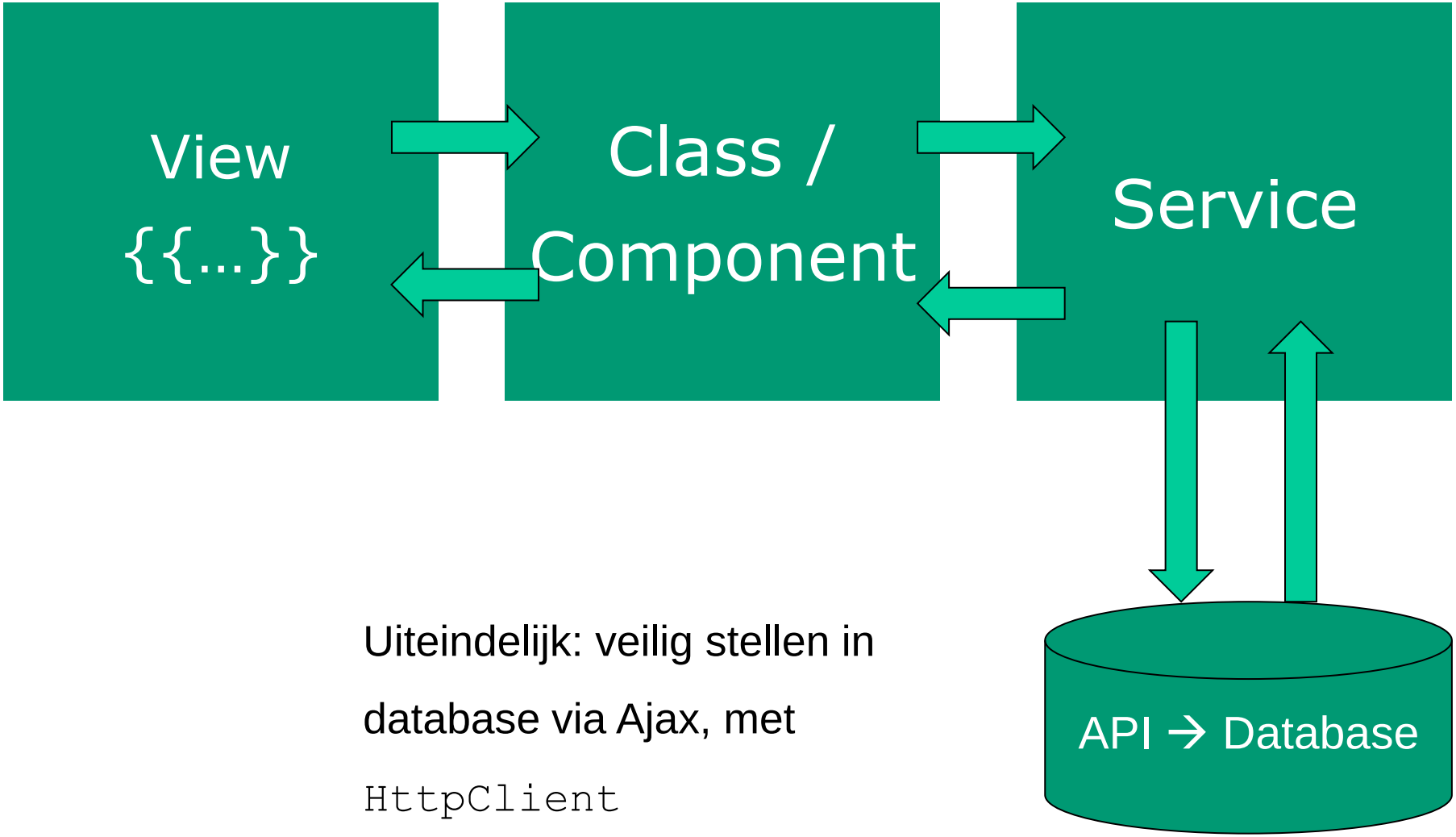
# Services

Doel – datafunctionality herbruikbaar maken voor verschillende componenten

- Data retrieval
- Data caching
- Data Storage,
- …

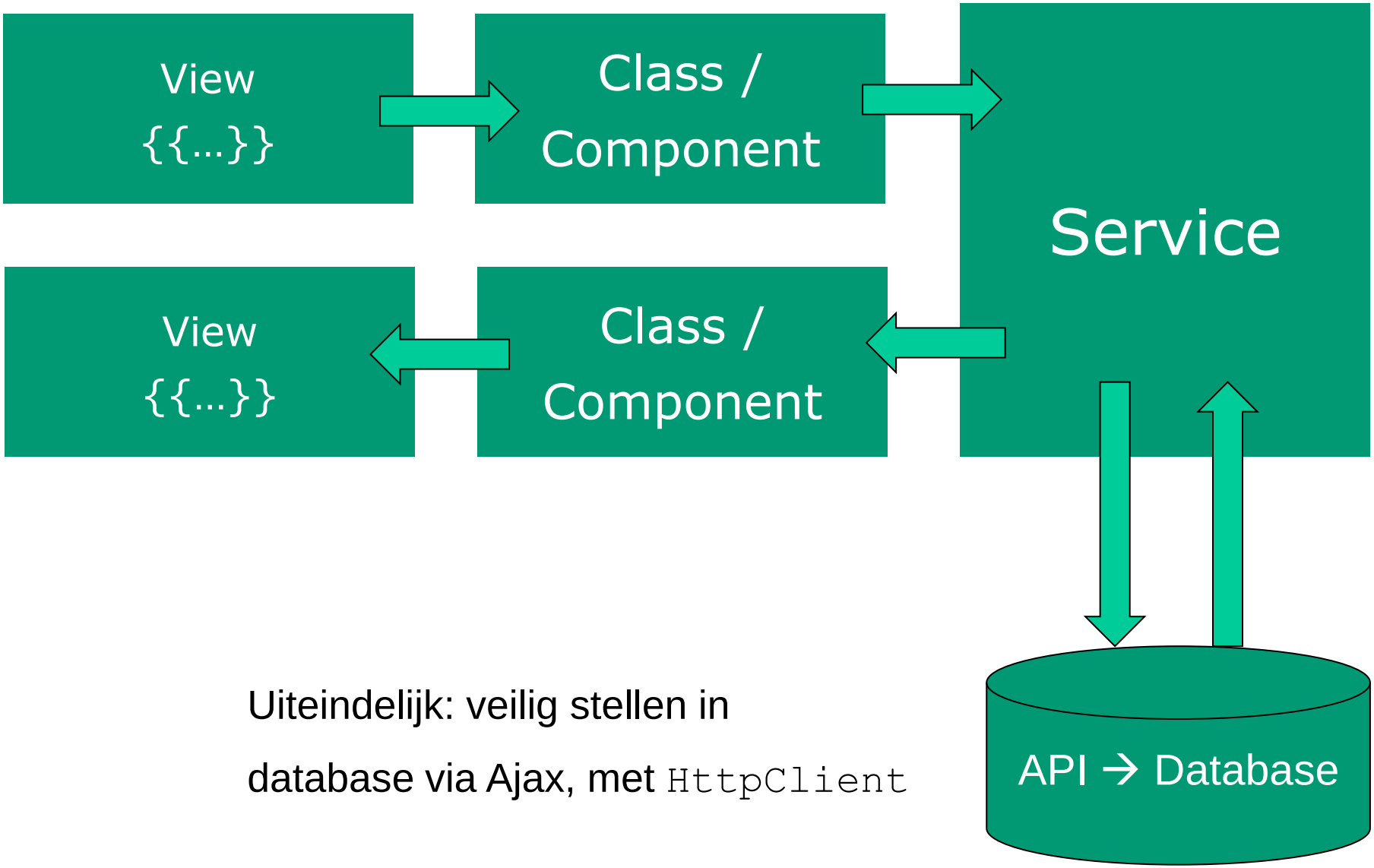
- Angular: één optie
  - `export class myDataService { … }`

# Data flow




Uiteindelijk: veilig stellen in database via Ajax, met `HttpClient`

# Data flow




Uiteindelijk: veilig stellen in database via Ajax, met `HttpClient`

# Services in Angular

Data services in Angular 1:

```
angular.module('myApp')
  .service(…)
  .factory(…)
  .provider(…)
```

Data services in Angular 2+:

```
import {Injectable} from '@angular/core';

@Injectable()
export class CityService{
   //....
}
```

# De rol van @Injectable

Why? – Dependency Injection (DI) en metadata!

*"TypeScript sees the @Injectable() decorator and emits metadata about our service, metadata that Angular may need to inject other dependencies into this service."*

https://angular.io/docs/ts/latest/tutorial/toh-pt4.html

*"Our service doesn't have any dependencies at the moment. Add the decorator anyway.*

*It is a best practice to apply the* `@Injectable()` *decorator* ***from the start*** *both for consistency and for future-proofing"*

# Creating a service

Creating a service in 3 steps

# Creating a service

```
ng generate service [name]
```

# Stap 1 – service maken (static data)

```
import { Injectable } from '@angular/core';
import { City } from './city.model'

@Injectable()
export class CityService {
   private cities:City[] = [
      new City(1, 'Groningen', 'Groningen'),
      …
   ];

   // retourneer alle cities
   getCities() {
      return this.cities
   }

   // retourneer city op basis van ID
   getCity(id:number) {
       return this.cities.find(c => c.id === id);
   }
}
```

# Stap 2 – Service consumeren/injecten

```
…
import {CityService} from "./city.service";

@Component({
   selector   : 'hello-world',
   templateUrl: 'app/app.component.html',
})

export class AppComponent implements OnInit {
   // Properties voor de component/class
   currentCity: City;
   cities: City[];
   cityPhoto: string;

   constructor(private cityService: CityServi

   }

   ngOnInit() {
      this.cities = this.cityService.getCities();
   }

   getCity(city: City) {
      this.currentCity = this.cityService.getCity(city.id);
      this.cityPhoto   = `img/${this.currentCity.name}.jpg`;
      console.log('City opgehaald:', this.currentCity);
   }
}
```

Constructor: DI + shorthand voor nieuwe private variable + instantiering!

local variables

Ophalen van de cities en lokale toekenning

Detailgegevens voor city bij (click) event

# Instantiation?

- Let op: geen `new()` instantie van de Service!
  - Services zijn Singletons
  - Worden opgehaald uit de Module en/*of* geïnstantieerd in een `constructor()`

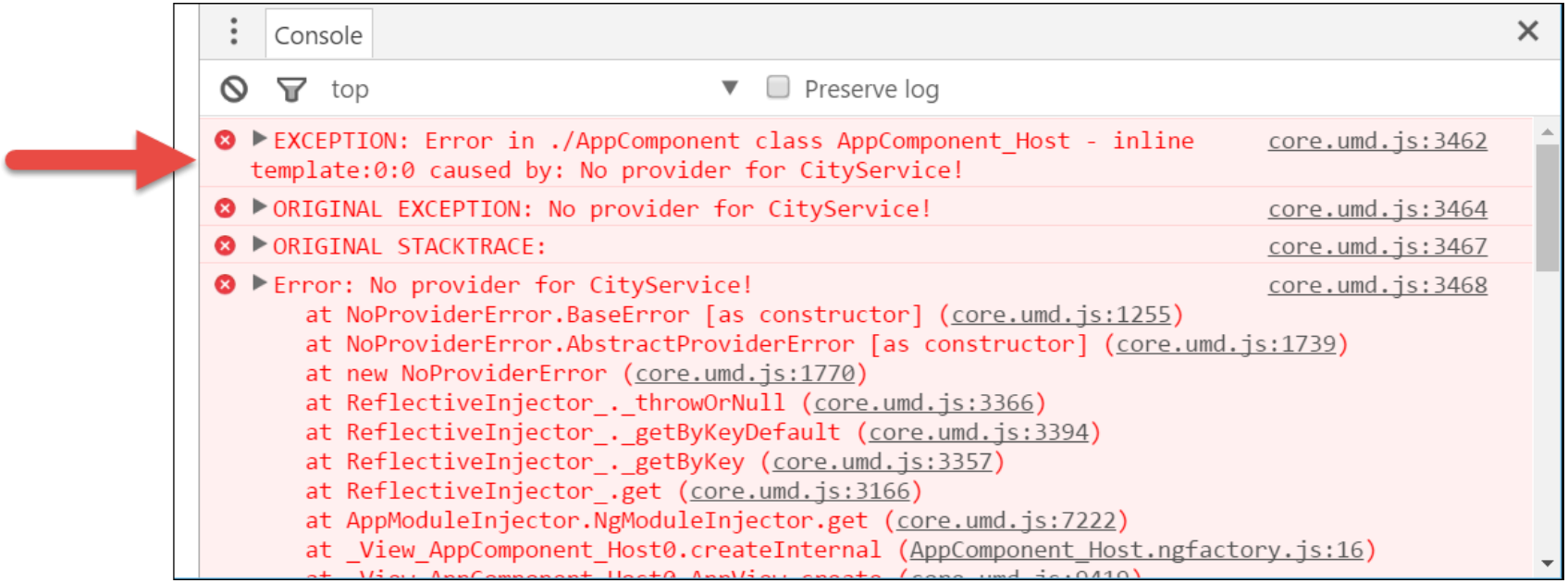
```
constructor(private cityService:CityService) { … }
```

*"The constructor itself does nothing.*

*The parameter simultaneously defines a private cityService property and identifies it as a CityService injection service."*

# “No provider for CityService”

- Solution: inject in `app.module.ts`



# Optie 1: Service injecteren in Module

- Alleen de *referentie* naar CityService is niet voldoende.
- Angular moet de service *injecteren* in de module
- Gebruik de annotatie `providers: [ … ]`

```
// Module declaration
@NgModule({
   imports      : [BrowserModule],
   declarations: [AppComponent],
   bootstrap    : [AppComponent],
   providers    : [CityService] // DI voor service
})
export class AppModule {
}
```

Array met Service-dependencies

# Optie 2 : Angular 6+, gebruik `providedIn`

- "Tree shakeable providers"
- Niet meer opgeven welke services in een Module worden gebruikt, maar andersom:
- In de service opgeven in welke modules deze wordt gebruikt.

```
@Injectable({
   providedIn: 'root'
})
export class CityService {
      …
}
```

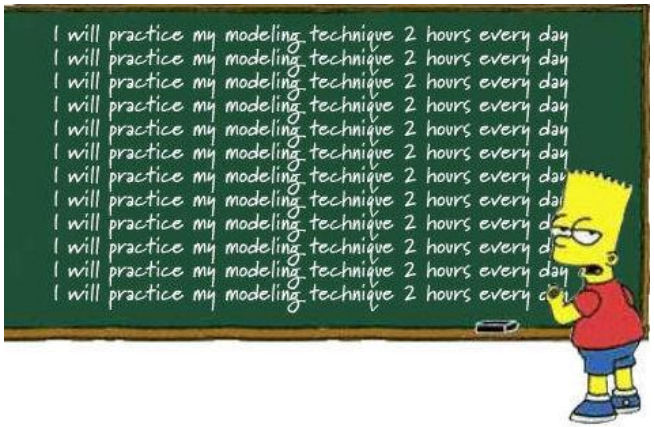
```
@NgModule({
   imports      : [BrowserModule],
   declarations: [AppComponent],
   bootstrap    : [AppComponent],
   // providers    : [CityService]
})
```

https://blog.angular.io/version-6-of-angular-now-available-cc56b0efa7a4

# Singleton?

- Services zijn (in principe) singletons
  - Maar: afhankelijk van de plek waar ze geïnstantieerd worden!
  - Ze zijn een singleton voor de module en alle child components.
  - Module/Site-wide gebruiken? Instantieer service in `app.module.ts` of gebruik `providedIn:'root'`

# Checkpoint

- Elke service in Angular 2 is een `class`
- Service importeren in de component waarin je hem injecteert
- Dependency Injection in constructor van `Component`
- Vergeet niet: Instantiëren in `ngModule` OF: `providedIn: 'root'` gebruiken
- Functies van de service gebruiken in de componenten
- Voorbeeld: `/200-services-static`
- Oefening `5a)`

**Workshop….**